

$$C_1 e^t$$

$$\frac{d^2 y}{dt^2} + m \frac{dy}{dt} + ny = 0$$

$$C_1 e^t + C_2 e^{-t} + m(C_1 e^t - C_2 e^{-t}) + n(C_1 e^t + C_2 e^{-t}) = 0$$

$$(C_1 e^t + C_2 e^{-t})(1+n) + m(C_1 e^t - C_2 e^{-t}) = 0$$

$$C_1 e^t(1+n+m) + C_2 e^{-t}(1+n-m) = 0$$

$$\Rightarrow m+n+1 = 0$$

$$m+n = -1$$